THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 1 & 2: **Chemical Reactions and Equations, Acids, Bases and Salts** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** all of these	All are signs of a chemical change
2	**(b)** $Fe_2O_3 \cdot x\,H_2O$	Hydrated iron(III) oxide
3	**(a)** conservation of mass	Mass is neither created nor destroyed
4	**(b)** yellow	Lead iodide is yellow
5	**(c)** Incorrect — melting is physical but burning is chemical	Both changes occur
6	**(a)** sour	Bases taste bitter
7	**(d)** base	Natural indicator
8	**(b)** curd	Sour milk
9	**(b)** Incorrect — it is neutralised by a base	Acid + base → neutral
10	**(c)** acidic	CO_2 reacts with $Ca(OH)_2$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	It reacts vigorously, releasing heat, forming slaked lime: $CaO + H_2O \longrightarrow Ca(OH)_2$	1+1
12	(i) $2\,H_2 + O_2 \longrightarrow 2\,H_2O$ (ii) $Mg + 2\,HCl \longrightarrow MgCl_2 + H_2$	1+1
13	Litmus (from lichen), turmeric, red cabbage, hydrangea/petunia flowers (any two)	1+1
14	Bases soluble in water, e.g. NaOH, KOH	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	(i) displacement — Cl displaces I (ii) decomposition (iii) combination	1+1+1
16	Forms salt and water: $Ca(OH)_2 + CO_2 \longrightarrow CaCO_3 + H_2O$ — so non-metal oxides are acidic.	1+1+1
17	(i) $Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$ (ii) $Na_2SO_4 + BaCl_2 \longrightarrow BaSO_4 + 2\,NaCl$ (iii) $2\,Mg + O_2 \longrightarrow 2\,MgO$	1+1+1
	OR: One substance oxidised and another reduced. (i) oxidising agent Fe_2O_3, reducing agent Al (ii) oxidising agent SO_2, reducing agent H_2S	1+1+1
18	HCl/H_2SO_4 solutions glow a bulb in a circuit (ions present); glucose/alcohol solutions do not — they do not give H^+.	2+1
	OR: (i) sodium carbonate decahydrate $Na_2CO_3 \cdot 10\,H_2O$ — cleaning (ii) sodium hydrogencarbonate $NaHCO_3$ — antacid (iii) calcium oxychloride $CaOCl_2$ — disinfecting water	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $Fe_2O_3 \cdot x\,H_2O$; air (oxygen) and moisture (ii) paint keeps air and water away (iii) galvanising, oiling/greasing (iv) huge losses — iron structures, bridges, ships get weakened	2+1+1+1
	OR: (i) Two or more substances combine to form one product (ii) $C + O_2 \longrightarrow CO_2$; $2\,H_2 + O_2 \longrightarrow 2\,H_2O$; $CaO + H_2O \longrightarrow Ca(OH)_2$ (iii) heat is released during them	1+3+1
20	(i) A < B < C < D (ii) B (iii) D (iv) A (v) B	1+1+1+1+1
	OR: (i) Ionic compounds formed by neutralisation; strong acid + weak base → acidic; weak acid + strong base → basic; strong + strong → neutral (ii) NaCl neutral; Na_2CO_3 basic; NH_4Cl acidic	2+3

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME — SET B

Chapters 1 & 2: **Chemical Reactions and Equations, Acids, Bases and Salts** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** displacement reaction	Na displaces H
2	**(a)** Student A is correct	CuO loses oxygen
3	**(d)** an exothermic reaction	Heat is released
4	**(a)** rancidity	They prevent oxidation
5	**(a)** aqueous solution	Dissolved in water
6	**(b)** Student A is correct	Ethanoic acid
7	**(c)** more than 7	Basic
8	**(d)** sodium hydroxide	Fully ionised
9	**(c)** CO_2	Lime water turns milky
10	**(c)** evaporation	Salt pans

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A reaction producing an insoluble solid (precipitate), e.g. $Na_2SO_4 + BaCl_2 \longrightarrow BaSO_4\downarrow + 2\,NaCl$	1+1
12	False — rancidity is due to oxidation of fats and oils by air.	2
13	NaCl is formed from a strong acid and a strong base, so it is neutral (pH 7).	2
14	False — it happens in daily life too, e.g. antacids neutralise stomach acid and baking soda neutralises a bee sting.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Corrosion (rusting of iron), rancidity (stale chips), combustion (burning of fuels releasing energy)	1+1+1
16	(i) they are mild bases neutralising excess HCl (ii) to neutralise acids formed in the mouth (iii) to neutralise soil acidity	1+1+1
17	(i) $2\,K + 2\,H_2O \longrightarrow 2\,KOH + H_2$ (ii) $ZnCO_3 \longrightarrow ZnO + CO_2$ (iii) $Ca(OH)_2 + CO_2 \longrightarrow CaCO_3 + H_2O$	1+1+1
	OR: Ions of the two reactants are exchanged: $NaOH + HCl \longrightarrow NaCl + H_2O$; Na^+ pairs with Cl^- and H^+ with OH^-.	1+1+1
18	Heat baking soda: $2\,NaHCO_3 \longrightarrow Na_2CO_3 + H_2O + CO_2$; recrystallise: $Na_2CO_3 + 10\,H_2O \longrightarrow Na_2CO_3\cdot 10\,H_2O$	1½+1½
	OR: H_3O^+ decreases; pH increases; the base neutralises the acid and finally OH^- is in excess.	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) (a) $BaCl_2(aq) + Na_2SO_4(aq) \longrightarrow BaSO_4(s) + 2\,NaCl(aq)$ (b) $NaOH(aq) + HCl(aq) \longrightarrow NaCl(aq) + H_2O(l)$ (c) $Zn(s) + H_2SO_4(aq) \longrightarrow ZnSO_4(aq) + H_2(g)$ (ii) double displacement, double displacement (neutralisation), displacement (iii) (a), $BaSO_4$	3+1+1
	OR: (i) dazzling white (ii) MgO; forms basic $Mg(OH)_2$ — turns red litmus blue (iii) $2\,Mg + O_2 \longrightarrow 2\,MgO$; $MgO + H_2O \longrightarrow Mg(OH)_2$ (iv) to remove the oxide layer	1+1+2+1
20	(i) $NaOH + HCl \longrightarrow NaCl + H_2O$ (ii) NaOH + phenolphthalein turns pink; adding HCl drop by drop makes it colourless; adding NaOH again restores pink (iii) neutralisation is exothermic	1+3+1
	OR: (i) $CaSO_4\cdot 2\,H_2O$ (ii) heating at 373 K (iii) $CaSO_4\cdot\frac{1}{2}\,H_2O + \frac{3}{2}\,H_2O \longrightarrow CaSO_4\cdot 2\,H_2O$ (iv) supporting fractured bones	1+1+2+1

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ANSWER KEY & MARKING SCHEME SET C

Chapters 1 & 2: **Chemical Reactions and Equations, Acids, Bases and Salts** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** $ZnO + C \longrightarrow Zn + CO$	ZnO reduced, C oxidised
2	**(c)** Incorrect — it is a double displacement (neutralisation) reaction	Ions are exchanged
3	**(d)** Fe_2O_3 and aluminium	$Fe_2O_3 + 2\,Al \longrightarrow Al_2O_3 + 2\,Fe$
4	**(a)** Student A is correct	$3\,Fe + 4\,H_2O \longrightarrow Fe_3O_4 + 4\,H_2$
5	**(b)** decomposition of calcium carbonate	Heat is absorbed
6	**(a)** 2	$CaSO_4 \cdot 2\,H_2O$
7	**(b)** Incorrect — when it is acid rain its pH is less than 5.6	Dissolved gases lower pH
8	**(c)** water of crystallisation	Anhydrous $CuSO_4$ is white
9	**(a)** $Mg(OH)_2$	A mild base
10	**(a)** Student A is correct	Na_2CO_3 is soda ash

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	It decomposes in light: $2\,H_2O_2 \longrightarrow 2\,H_2O + O_2$; dark bottles prevent this.	1+1
12	An unbalanced equation, e.g. $Mg + O_2 \longrightarrow MgO$	1+1
13	Black CuO dissolves forming blue-green copper chloride: $CuO + 2\,HCl \longrightarrow CuCl_2 + H_2O$	1+1
14	(i) tartaric acid (ii) citric acid	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Displacement: a more reactive element displaces another, e.g. $Fe + CuSO_4 \longrightarrow FeSO_4 + Cu$. Double displacement: exchange of ions, e.g. $AgNO_3 + NaCl \longrightarrow AgCl + NaNO_3$.	1+1+1
16	Distilled water has almost no ions; rain water contains dissolved acids/salts which provide ions.	3
17	M = iron; oxide Fe_3O_4; $3\,Fe + 4\,H_2O \longrightarrow Fe_3O_4 + 4\,H_2$	1+1+1
	OR: Fe displaces Cu forming green $FeSO_4$. No reaction with Cu in $FeSO_4$, because copper is less reactive than iron.	1+1+1
18	A — HCl is a strong acid with more H^+ ions than acetic acid (weak acid).	1+2
	OR: Cooking/preserving food; raw material for NaOH, baking soda, washing soda, bleaching powder; de-icing roads	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) oxidation: gain of O/loss of H; reduction: loss of O/gain of H (ii) H_2 oxidised, CuO reduced; C oxidised, ZnO reduced (iii) a substance that supplies oxygen/removes hydrogen, itself getting reduced	2+2+1
	OR: (i) oxidation of fats/oils giving bad smell and taste (ii) nitrogen is unreactive and removes oxygen (iii) substances that prevent oxidation (iv) keeps out air (v) slows down the oxidation reaction	1+1+1+1+1
20	(i) SO_2 and NO_2 (ii) less than 5.6 (iii) lowers pH of water bodies, killing fish (iv) corrodes marble ($CaCO_3$) (v) use cleaner fuels / scrubbers	1+1+1+1+1
	OR: (i) salt + water + CO_2 (ii) $CaCO_3 + 2\,HCl \longrightarrow CaCl_2 + H_2O + CO_2$; $NaHCO_3 + HCl \longrightarrow NaCl + H_2O + CO_2$ (iii) passing it through lime water turns it milky	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET D

Chapters 1 & 2: **Chemical Reactions and Equations, Acids, Bases and Salts** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** silver bromide	Light-sensitive
2	**(b)** oxygen	Anode is positive
3	**(d)** oxidised	Gains oxygen
4	**(c)** Incorrect — mass is conserved; it appears less only if gases escape an open system	Law of conservation of mass
5	**(d)** hydrogen	$Zn + 2\,HCl \longrightarrow ZnCl_2 + H_2$
6	**(a)** strongly basic	Near 14
7	**(a)** methyl orange	Synthetic indicator
8	**(d)** methanoic acid	Formic acid
9	**(d)** NH_4OH and HCl	Base + acid
10	**(a)** Correct	Definition

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A yellow precipitate of lead iodide forms; the solution remains colourless. $Pb(NO_3)_2 + 2\,KI \longrightarrow PbI_2 + 2\,KNO_3$	1+1
12	It forms a thin, tough layer of aluminium oxide that protects it from further reaction.	2
13	Salts with the same positive or negative radical, e.g. NaCl and Na_2SO_4 (sodium salts)	1+1
14	It has fewer H_3O^+ ions per unit volume.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Gas bubbles, test tube becomes warm, zinc dissolves gradually; $Zn + 2\,HCl \longrightarrow ZnCl_2 + H_2$	2+1
16	Hydrated salts contain water of crystallisation ($CuSO_4 \cdot 5\,H_2O$); anhydrous salts do not ($CuSO_4$).	1+1+1
17	Light supplies energy for decomposition: $2\,AgBr \xrightarrow{\text{light}} 2\,Ag + Br_2$; used in black-and-white photography.	1+1+1
	OR: (i) $Na_2CO_3 + 2\,HCl \longrightarrow 2\,NaCl + H_2O + CO_2$ — double displacement (ii) $2\,KClO_3 \longrightarrow 2\,KCl + 3\,O_2$ — decomposition (iii) $2\,Fe + 3\,Cl_2 \longrightarrow 2\,FeCl_3$ — combination	1+1+1
18	Stomach HCl aids digestion; excess acid causes pain and irritation; antacids (mild bases like milk of magnesia) neutralise it.	1+1+1
	OR: (i) $Fe + 2\,HCl \longrightarrow FeCl_2 + H_2$ (ii) effervescence of CO_2 (iii) solution stays colourless until neutralised, then turns pink with excess base	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) to obey the law of conservation of mass (ii) (a) $3\,Fe + 4\,H_2O \longrightarrow Fe_3O_4 + 4\,H_2$ (b) $C_3H_8 + 5\,O_2 \longrightarrow 3\,CO_2 + 4\,H_2O$ (c) $4\,NH_3 + 5\,O_2 \longrightarrow 4\,NO + 6\,H_2O$	2+3
	OR: (i) carbon electrodes in water with a little dilute H_2SO_4, test tubes over electrodes, battery (ii) H_2 at cathode (pop sound), O_2 at anode (rekindles a glowing splinter) (iii) 2 : 1, since water is H_2O (iv) decomposition by electricity	2+1+1+1
20	(i) (a) about 1.2 (b) 7 (c) about 10 (ii) living organisms survive only in a narrow pH range (7–7.8) (iii) makes soil acidic, unsuitable for plant growth	3+1+1
	OR: (i) strong: fully ionised (NaOH); weak: partially (NH_4OH) (ii) (a) it gives Na^+ and OH^-, releasing heat (b) OH^- concentration per unit volume decreases (iii) highly exothermic — may splash and burn	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.